Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional purchase data to identify trends in customer demographics, purchasing patterns, product preferences, payment methods, and shopping frequency. The objective is to transform raw retail transaction data into actionable business insights using **Python, PostgreSQL, and Power BI**. The findings can help businesses improve marketing strategies, optimize inventory, enhance customer retention, and increase overall sales performance.

2. Dataset Summary

Dataset Information

- **Dataset Name:** Customer Shopping Behavior
- **Rows:** 3,900
- **Columns:** 18

Key Features

Customer Information

- Customer ID
- Age
- Gender
- Location

Product Information

- Item Purchased
- Category
- Size
- Color
- Season

Transaction Details

- Purchase Amount (USD)
- Shipping Type
- Payment Method

- Discount Applied
- Promo Code Used
- Subscription Status

Customer Behaviour

- Previous Purchases
- Frequency of Purchases
- Review Rating

Data Quality

- Total Records: **3,900**
- Total Features: **18**
- Missing Values: **Review Rating contains missing values**
- Duplicate Records: **None**
- Numeric Features: Age, Purchase Amount, Review Rating, Previous Purchases
- Categorical Features: Remaining columns

3. Exploratory Data Analysis using Python

The dataset was cleaned, explored, and transformed using **Python (Pandas, NumPy, Matplotlib, and Seaborn).**

Data Loading

- Imported dataset using Pandas.
- Loaded CSV into a DataFrame.

Initial Exploration

- Checked dataset dimensions.
- Examined data types.
- Generated descriptive statistics.
- Identified numerical and categorical columns.

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object 
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
data.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN

Data Cleaning

- Checked for duplicate records.
- Checked missing values.
- Filled missing Review Rating values using the median.
- Verified data consistency.

Feature Engineering

- Created **Age Group** categories.
- Standardized column names to snake_case.
- Converted categorical variables where required.
- Prepared data for SQL and Power BI.

```
labels=['Young Adult','Adult','Middle-aged','Senior']
data['age_group']=pd.qcut(data['age'],q=4,labels=labels)
```

```
data[['age','age_group']]
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
...	...	...
3895	40	Adult
3896	52	Middle-aged
3897	46	Middle-aged
3898	44	Adult
3899	52	Middle-aged

3900 rows × 2 columns

```
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
data['purchase_frequency_days']=data['frequency_of_purchases'].map(frequency_mapping)
```

```
data[['frequency_of_purchases','purchase_frequency_days']]
```

	frequency_of_purchases	purchase_frequency_days
0	Fortnightly	14
1	Fortnightly	14
2	Weekly	7
3	Weekly	7
4	Annually	365
...	...	...
3895	Weekly	7
3896	Bi-Weekly	14
3897	Quarterly	90
3898	Weekly	7

Database Integration

- Connected Python with PostgreSQL.
- Imported the cleaned dataset into PostgreSQL using SQLAlchemy.
- Verified successful table creation for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

The cleaned data was imported into PostgreSQL to answer important business questions.

SQL Analysis Performed

1. Total Revenue by Product Category
2. Revenue by Gender
3. Average Purchase Amount by Age Group
4. Top 10 Highest Selling Products
5. Most Frequently Purchased Product Categories
6. Revenue by Season
7. Customer Spending by Subscription Status
8. Payment Method Usage Analysis
9. Shipping Type Performance
10. Customers with Highest Purchase Amount
11. Discount vs Non-Discount Purchase Comparison
12. Average Review Rating by Category
13. Top Locations by Revenue
14. Purchase Frequency Analysis
15. Previous Purchases vs Current Spending
16. Revenue by Color Preference
17. Revenue by Size
18. Seasonal Product Performance
19. Top Spending Customers
20. Customer Segmentation based on Previous Purchases

5. Dashboard in Power BI

An interactive Power BI dashboard was developed to visualize customer purchasing behavior.

Dashboard KPIs

- Total Customers
- Total Revenue
- Average Purchase Amount
- Average Review Rating
- Total Previous Purchases

Interactive Filters

- Gender
- Category
- Season
- Shipping Type
- Subscription Status
- Payment Method
- Location

Dashboard Visualizations

- Revenue by Category
- Revenue by Season
- Purchase Amount by Age Group
- Sales by Gender
- Subscription Status Distribution
- Payment Method Distribution
- Shipping Type Comparison
- Top Selling Products
- Revenue by Location

- Previous Purchases Analysis

6. Business Recommendations

Increase Customer Loyalty

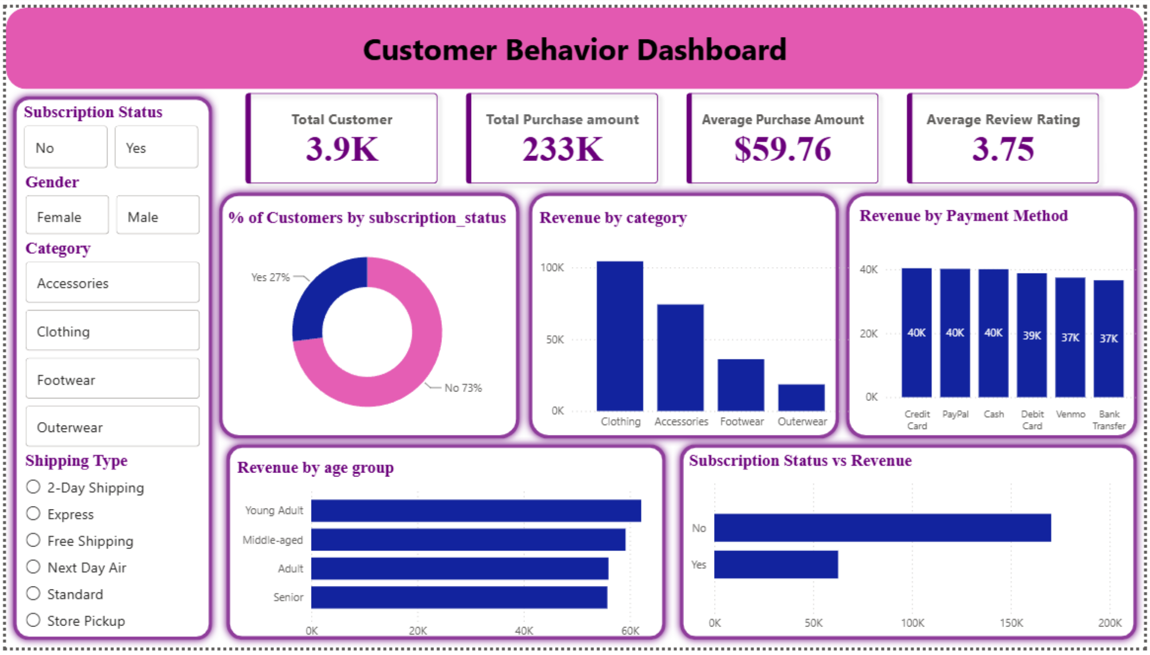
Introduce personalized rewards and loyalty programs for repeat customers with high previous purchase counts.

Target High-Revenue Categories

Invest more in categories contributing the highest revenue through inventory expansion and promotional campaigns.

Seasonal Marketing

Launch targeted promotions during peak-performing seasons to maximize sales.



Optimize Discount Strategy

Evaluate discount effectiveness to ensure promotions increase profitability rather than only transaction volume.

Improve Subscription Adoption

Provide exclusive offers and early access benefits to encourage more customers to subscribe.

Focus on High-Value Customers

Identify customers with consistently high purchase amounts and offer personalized recommendations and premium membership benefits.

Enhance Customer Experience

Monitor products receiving lower review ratings and improve product quality or customer service.

Optimize Shipping Options

Promote shipping methods that balance customer satisfaction and operational costs.